



Srishyla Educational Trust (R)

GM INSTITUTE OF TECHNOLOGY

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Member of
NASSCOM

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

Subject: Introduction to Electronics Engineering

Sem: I 'B'

Assignment Questions : Module 2

Part A

1. Explain the pin configuration of an OPAMP. Mention the OPAMP ideal characteristics.
2. Explain the OPAMP parameters
3. Draw neat circuit diagram, input output waveforms and output voltage expressions of the following OPAMPs i) inverting amplifier, ii) non-inverting amplifier, iii) differentiator iv) integrator and v) summing amplifier.
4. Explain the following with a neat diagram
a) comparator (b) a differentiator (c) an integrator (d) Inverting Amplifier e) Voltage follower

Part B

1. Explain the criteria for oscillations.
2. Define positive feedback. With the neat diagram derive an expression for overall gain with positive feedback
3. Explain the operation of a three-stage ladder RC network oscillator and Wein bridge oscillator.
4. Define multivibrator. Explain the different types of multivibrator
5. With suitable diagrams explain single stage astable multi-vibrator using operational amplifier.
6. Define sinusoidal oscillator, non sinusoidal oscillator, crystal controlled oscillator
7. Determine the frequency of oscillation of a three-stage ladder RC network oscillator in which $C = 10 \text{ nF}$ and $R = 10 \text{ k}\Omega$.
8. The frequency sensitivity arms of the Wein bridge oscillator uses $C_1 = C_2 = 0.01 \mu\text{F}$ and $R_1 = 10 \text{ k}\Omega$ while R_2 is kept variable. The frequency is to be varied from 10 KHz to 50 KHz by varying R_2 . Find the minimum and maximum values of R_2 .